

Derived-Operator Dynamics of a Catalytic-Domain Contact Network

long-range contact response at the causal frontier, measured on 156 qubits

Answering the challenge statement section by section · every number carries a cloud job receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-cleveland-gic2026>

Executive summary

8.68 σ

long-range contact response $C(28-7) = -0.0757 \pm 0.0087$ at lattice distance 11, measured on **ibm_fez**, **156 qubits**, K=6 native-edge Trotter steps — a parameter-free force graph, nothing fitted

99 qubits

the exact lightcone for this weight-2 observable at this depth. The correlation is carried by a region no pairwise contact heuristic represents

\$0.2–2M

per campaign from variant triage at one-in-five deflection on a 200-variant programme, wet-lab cost alone. No affinity claim; no replacement of assay

Why the numbers can be trusted

The force graph is parameter-free. Site energies come from Coulomb terms plus a 0.21 eV hydrogen-bond scale, referenced to the thermal floor kT ; the coupling $\kappa = 3/(3-\sqrt{5})$ is derived from the model's structure. Nothing is fitted to the contact map being measured, so there is no way for the answer to be smuggled into the input. A credential-free audit (`python verify.py`) replays **47 checks** against archived counts with no token required.

The headline is a pre-registered hardware measurement, not a simulation. $C(28-7) = -0.0757 \pm 0.0087$ is a **8.68 σ** long-range contact response across lattice distance 11 on **ibm_fez** at 156 qubits, after six native-edge Trotter steps. The exact lightcone for that observable spans **99 qubits**, so the correlation is carried by a region no pairwise contact heuristic represents. Negative and null arms ship at the same receipt class as the positive one.

Technical report follows (\$0–\$8) · Reproducibility & Receipts companion document supplied separately · <https://github.com/sharadbachani-oss/merlin-quantum-cleveland-gic2026>

Technical Report

Merlin Digital — Cleveland Clinic track, package v4. Every number below was re-derived from `RESULT_PREREG` in `verify.py`. Job ids appear only when they are stored in an artifact.

1. Claim

A parameter-free force graph — Coulomb at derived α plus the 0.21 eV H-bond chemistry pair, with site energies on the derived energy ladder at the thermal floor kT — was quenched on today's hardware. On **ibm_fez**, 156 qubits, $K=6$ native-edge Trotter steps:

$> C(28-7) = -0.0757 \pm 0.0087 \text{ (8.68}\sigma\text{)}$ at lattice distance 11.

That pair sits at the causal frontier of the evolution. The exact lightcone for a weight-2 observable at this depth spans 99 qubits. No classical prediction was registered for $K=6$; the $K=2$ card on the same layout was checked against an exact cone first.

The same derived U feeds a mediated-response quadratic $(I-GU)x = e_{\text{act}}$, solved on QCi Dirac-3 for all five challenge targets, and an analog layout on QuEra Aquila whose atom positions are the $C\alpha$ coordinates.

2. Physics (our first-principles engine L0–L2 only)

See `PHYSICS.md`. One operator:

```
`` H = κ · D - A■ , κ = 3/(3-√5), δ = 0.839229, E■ = -2.517687 ``
```

Exact shallow prep: $\theta^* = \pi/2 - 2 \operatorname{atan}(\delta/2)$, three CNOTs, one per commuting coupled qubit pair. Parent the derived-operator model v4+ biology layers (HexaGene, codon/amino overlays, genome axes) are out of this package. Residue identities enter as molecular masses and atomic coordinates.

The 11-cell card measures $U/kT = 106.64$. That is the strong-coupling regime: no small parameter, collective modes. Perturbative / mean-field classics are invalid *by construction* for the connected correlator. That is the advantage argument's first leg — a regime claim, not a runtime claim.

3. Hardware that ran

3.1 Checkable gate-model (trust the instrument)

R1 — 11 cells, ibm_marrakesh, EstimatorV2 res-2, 166 two-qubit gates. $C(1-0) = +0.6805 \pm 0.0234$ vs $+0.7421$ exact. 3/3 signal pairs in band. Job `daau17bvpcac73dd29r0` recovered 2026-09-08 by exact evs match.

R2 — 156 qubits, ibm_fez, K=2. $C(6-7) = -0.4584 \pm 0.0076$ vs -0.4648 exact lightcone (cone size 6). In band. Backend in the `RESULT` file is **ibm_fez**. An earlier draft listed `marrakesh`; that is corrected here. Job `dab6vi34clkc73fij1ig`.

3.2 Advantage card (unverified width, nulls on)

R3 — 156 qubits, ibm_fez, K=6, 694 two-qubit gates after transpile. 120 pre-registered pairs. Headline $C(28-7)$ at $d=11$ is 8.68σ from zero. Supporting structure at neighbouring distances (see fig1). No exact prediction exists at this depth; the $K=2$ cone on the same assignment is the checkable sibling. Job `dab7i5mrr17c7386gfh0`.

This is a **lightcone-dimension** bound (2^{99}), not a measured MPS/operator death of the Mitsubishi class. The package says so (`QUANTUM_ADVANTAGE_EXHIBIT.md`).

3.3 Dirac-3 mediated response — five targets, real job ids

Target	vars	cosine	obj gap	G1	G2	job id
KRAS G12C	339	0.9973	0.77%	PASS	PASS	6a96810808442f441bbb6b2d
Glucokinase	849	0.9943	2.1%	PASS	PASS	6a96a04d08442f441bbb6c85
CDK2	589	0.9649	6.9%	PASS	PASS	6a96a36208442f441bbb6c87

Target	vars	cosine	obj gap	G1	G2	job id
BCR-ABL1	581	0.9684	5.0%	PASS	PASS	6a96a50f08442f441bbb6c8b
Cardiac myosin (pass 1)	949	0.9267	14.0%	FAIL	PASS	6a96af7408442f441bbb6c8f
Cardiac myosin (refine)	949	0.9916	1.6%	PASS	PASS	6a96c15d08442f441bbb6c91

Myosin pass 1 hit the device floor; a pre-registered refine closed G1. Do not quote “five of five PASS” on pass 1.

3.4 Aquila — geometry is the Hamiltonian

T0 (10 atoms): 7/7 signal keys in band, 7/7 signs. Strongest C(6-5) = -0.12247 vs -0.12328 predicted (0.66%). Tasks: aws:quera:qpu:aquila-5879-qjob-6a9a83b2e026787d1e5b06d0 (T0), ...6a9a83b3e026787d1e5b06d3 (T1, 60 atoms). T1 front: near ($<12\text{ }\mu\text{m}$) mean $|C|$ 0.0157 vs far ($\geq 30\text{ }\mu\text{m}$) 0.0105, ratio 1.50.

3.5 Rigetti Cepheus-1 — structure, not amplitude

Job openquantum:rigetti:qpu:cephus-1-108q-5879-qjob-6a99bf4ce026787d1e5ae579. Pearson +0.769 vs exact lightcone; 33/37 sign agreement; 62 nulls silent. **Amplitude gate FAILED** (slope 0.378; ~16 dead qubits). Cross-vendor structure only.

4. What did not run

Exact protocols, unmarked jobs: PROTOCOL_PENDING.md.

- **IonQ Forte** 11-cell / 288 native 2q quench — cancelled after 30 h stall. UNRUN. One command when the queue is live: python protocol/ionq_quench.py fly.
- **AQT ibex-q1** ...6a99b67ae026787d1e5ae3d5 under sha 82dbe4ff... completed 2026-09-04. Harvested 512 shots. Grade **FAIL** (C(1-0)=+0.077 vs +0.768).
- **IBM residue K=6** for BCR-ABL, myosin, glucokinase, CDK2 — quota-gated. UNRUN. One command: python protocol/ibm_four_maps.py fly6 TARGET.

Commercial value

The measured object is a **long-range contact response** in a catalytic domain — the correlation between residues far apart on the contact graph, at the causal frontier of the evolution. That is the quantity that decides whether a distal mutation changes activity, which is the expensive question in protein engineering and allosteric drug design.

input	value	source
pre-clinical discovery cost per programme	\$0.5–1.0B	industry cost-of-capital studies
hit-to-lead design cycles	10–50 iterations	medicinal chemistry practice
wet-lab cost per designed variant	\$5–50k	expression + assay
object delivered	C(28–7) = -0.0757 ± 0.0087 (8.68 σ) at lattice distance 11	this package, ibm_fez

Priced as **variant triage**: a distal-coupling predictor that removes one in five variants from a 200-variant campaign saves **\$0.2–2M per campaign** in wet-lab cost alone, and more in cycle time. The exact lightcone for this observable spans 99 qubits — the correlation is carried by a region no pairwise contact heuristic represents.

What we do not claim. No binding-affinity prediction, no replacement of assay. This is a structural-response instrument; the affinity work in the annex is reported with its own gates and its own negatives.

5. Noise as instrument

R1 uses the production EstimatorV2 stack (twirl + TREX + multi-scale ZNE). The method ladder that produced it — routing inflation 288→757 CX, bare amplitude damping, global-echo overcorrect, two-point ZNE undershoot — is in the source tree and is **not** re-sold as a result. Aquila drops shots with empty sites. Dirac grades cosine and objective against a frozen exact vector, not a fitted landscape.

6. What Cleveland Clinic can use now vs after the remaining cards

Now: a KRAS communication front on a 156-qubit native map, a five-target response field on Dirac-3, and an analog geometry check. After IonQ + four IBM maps: a cross-vendor and cross-target gate-model set at the same depth. Until those RESULT files exist, this is a complete **protocol package**, not a finished send.

7. Figures

- `figures/fig1_k6_front.png` — $K=6$ $|C|$ vs lattice distance
- `figures/fig2_dirac_cosine.png` — five-target pass-1 cosines
- `figures/fig3_aquila_t0.png` — T0 pred vs hardware
- `figures/fig4_ran_vs_protocol.png` — honest status board